Chines Zodiac

The program determine the corresponding zodiac animal using the remainder of the user's birth year divided by 12. After user input their birth year, it will show which zodiac animal user's birth year is and initiate to ask user's birth year again.

Zodiac Cycle :

0 => Monkey
1 => Rooster
2 => Dog
3 => Pig
4 => Rat
5 => Ox
6 => Tiger
7 => Rabbit
8 => Dragon
9 => Snake
10 => Horse
11 => Sheep

Codes implementation:
The program was written in Python using Visual Studio Code (VS Code) as the code editor

```
while True:
    birth_year = int(input("Enter your birth year: "))
    i = birth_year % 12
    if i == 0:
        print("Your Chinese zodiac sign is: Monkey")
    elif i == 1:
        print("Your Chinese zodiac sign is: Rooster")
    elif i == 2:
        print("Your Chinese zodiac sign is: Dog")
    elif i == 3:
        print("Your Chinese zodiac sign is: Pig")
    elif i == 4:
        print("Your Chinese zodiac sign is: Rat")
    elif i == 5:
        print("Your Chinese zodiac sign is: Ox")
    elif i == 6:
        print("Your Chinese zodiac sign is: Tiger")
    elif i == 7:
        print("Your Chinese zodiac sign is: Rabbit")
```

```
    elif i == 8:
        print("Your Chinese zodiac sign is: Dragon")
    elif i == 9:
        print("Your Chinese zodiac sign is: Snake")
    elif i == 10:
        print("Your Chinese zodiac sign is: Horse")
    elif i == 11:
        print("Your Chinese zodiac sign is: Sheep")
```

The program checks the remainder of the user's birth year when divided by 12. If the remainder is equal to 0, 'Monkey' will be displayed as the user's Chinese zodiac sign. Otherwise, the program will proceed to evaluate the remaining conditions using elif statements. Each zodiac animal is assigned a specific remainder value based on the Chinese zodiac cycle.

Instead of using multiple if, elif, and else statements, we can use a Python dictionary to map each year's remainder (when divided by 12) to its corresponding Chinese zodiac animal. This makes the code shorter, cleaner, and easier to maintain.

```
while True:
    birth_year = int(input("Enter your birth year: "))
    i = birth_year % 12
    zodiac_signs = {
        0: "Monkey",
        1: "Rooster",
        2: "Dog",
        3: "Pig",
        4: "Rat",
        5: "Ox",
        6: "Tiger",
        7: "Rabbit",
        8: "Dragon",
        9: "Snake",
        10: "Horse",
        11: "Sheep"
    }
    print(f"Your Chinese zodiac sign is: {zodiac_signs[i]}")
```

The program was written in Python using Visual Studio Code as the code editor.

Example output :

```
Enter your birth year: 2007
Your Chinese zodiac sign is: Pig
```

By using a while loop, the program continuously repeats its execution, automatically returning to the beginning after completing each iteration. This allows users to enter another birth year without having to restart the program manually.

```
Enter your birth year:
```